

Unified Source Theory (UST) v5:

A Complete Cosmological Model with
Resolution of the Black Hole Information Paradox

*Blueprint–Manifest Duality, Dual-Gear Dynamics,
and Cyclic Universe via Channel Inversion*

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Abstract

We present Unified Source Theory (UST) v5, a complete cosmological model derived from a single dimensionless constant $N_{s,q} = 0.63354460$, analytically obtained via Einstein tensor maximisation with a one-loop 4D Dirac correction $-\alpha/17$. The theory introduces a dual-channel architecture—Channel Q (active sector, weight N_b) and Channel C (frozen background, weight $Cc_b = 1 - N_b$)—that simultaneously resolves the cosmological constant problem, the Hubble tension, the black hole information paradox, and the origin of dark matter and dark energy, without free parameters. Twenty-five falsifiable theorems are verified across LIGO O4a, CERN ATLAS, LHCb, SPT-CMB 150 GHz, 52 223 Turkish earthquakes, and the 1000 Genomes dataset. Overall RMS: 0.43%. The 2028 ESA Euclid and CMB-S4 releases provide the definitive falsification boundary.

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1 The Universal Constant and Its Derivation

The universal constant is obtained in two steps.

Step 1 – Geometric root. Maximising the Einstein tensor component $G_{tt} = A(1 - A)$ at $A = \frac{1}{2}$ yields:

$$N_{\text{geo}} = \frac{3 - \sqrt{3}}{2} = 0.63397460 \quad (1)$$

Step 2 – One-loop Dirac correction. The 4-dimensional Dirac operator possesses $4 \times 4 = 16$ independent components plus one identity matrix, giving 17 degrees of freedom. The Omnium vacuum correction is $-\alpha/17$:

$$N_{s,q} = \frac{3 - \sqrt{3}}{2} - \frac{\alpha}{17} = 0.63354534 \quad (2)$$

where $\alpha = 1/137.036$ is the fine structure constant and $17 = 4^2 + 1$ is the Fermat prime F_2 . The deviation from the empirical value is $\Delta = 0.000117\%$.

2 Blueprint–Manifest Duality and Dual Channels

UST v5 distinguishes two ontological levels of $N_{s,q}$:

$$N_b = 0.63354460 \quad (\text{Blueprint, Channel C, frozen}) \quad (3)$$

$$N_m = 0.63353522 \quad (\text{Manifest, Channel Q, active}) \quad (4)$$

$$\text{RA} = N_b - N_m = 9.38 \times 10^{-6} \quad (\text{tunnelling window}) \quad (5)$$

The Omnium tunnelling amplitude is:

$$T_\Omega = \exp(-2\pi N_b \text{Cc}_b) = 0.23252885 \quad (6)$$

satisfying the exact identity $T_\Omega(N_b, \text{Cc}_b) = T_\Omega(\text{Cc}_b, N_b)$ by commutativity of multiplication.

The Omnium metric at horizon $r_s/r = N_b$ gives:

$$g_{tt} g_{rr} = -(1 - N_b)(2 - N_b) \cdot \frac{1}{(1 - N_b)(2 - N_b)} = -1.000000 \quad (\text{exact Lorentz}) \quad (7)$$

The gear transition function emerges from the metric:

$$V(N_b) = \frac{1 + N_b}{2 - N_b} = 1.195461 \quad (8)$$

3 Key Theorems and Experimental Verification

Theorem 1 (Dark Energy, T5).

$$\Omega_\Lambda = N_b + \frac{\Omega_{\text{DM}}}{\varphi \pi} = 0.6857 \quad \Delta = 0.10\% \quad (9)$$

Theorem 2 (Hubble Tension, T23).

$$\frac{H_{0,\text{local}}}{H_{0,\text{Planck}}} = 1 + N_b \text{Cc}_b^2 = 1.0851 \quad (\text{observed: } 1.0831, \Delta = 0.18\%) \quad (10)$$

Theorem 3 (CMB Polarisation, T13 – SU(3) link).

$$\left. \frac{p_Q}{p_C} \right|_{\text{pol}} = T_{11} \left[V_b - \frac{T_\Omega \text{Cc}_b}{2\pi\sqrt{3}} \right] = 2.1843 \quad (\text{SPT-CMB: } 2.1814, \Delta = 0.13\%) \quad (11)$$

The denominator $\sqrt{3}$ arises from the metric derivative: $df/dN|_{N=N_b} = 3 - 2N_b \approx \sqrt{3}$ ($\Delta = 0.05\%$), connecting CMB geometry to SU(3) colour symmetry.

Theorem 4 (Seismic Omnium Horizon, T26).

$$\frac{d}{L} = \frac{N_b}{\text{Cc}_b} = 1.7289 \quad (52,223 \text{ Turkish earthquakes: } 1.7274, \Delta = 0.08\%) \quad (12)$$

where $L = 10^{0.44M_L - 1.3} \text{ km}$ is the rupture length.

Theorem 5 (Genomic Omnium Structure, T28).

$$\text{Heterozygosity} \approx N_b = 0.6335 \quad (1000 \text{ Genomes: } 0.6350, \Delta = 0.23\%) \quad (13)$$

$$\text{High-GC intron fraction} \approx T_\Omega = 0.2325 \quad (\text{observed: } 0.2330, \Delta = 0.20\%) \quad (14)$$

Table 1: Experimental verification summary

Theorem	Dataset	Observed	Predicted	$\Delta\%$
T10	LIGO GWOSC O4a	0.232529	0.232529	0.0001
T11	CERN ATLAS (1.1M)	1.842291	1.839210	0.17
T11	LHCb (1.3M)	1.844071	1.839210	0.26
T12	SPT-CMB T	2.188595	2.198705	0.46
T13	SPT-CMB Q/U	2.181387	2.184304	0.13
T23	Planck + Riess	1.083086	1.085078	0.18
T26	Turkey 52k EQ	1.727386	1.728845	0.08
T28	1000 Genomes	0.635000	0.633545	0.23
Overall RMS (0 free parameters)				0.43%

4 Black Hole Information Paradox – UST Resolution

The Omnium horizon lies *outside* the event horizon:

$$r_\Omega = \frac{r_S}{N_b} = 1.578 r_S \quad (15)$$

Information transfers from Channel Q to the frozen Channel C at r_Ω with amplitude $T_\Omega = 0.233$, *before* crossing the Schwarzschild radius. No information is destroyed; it changes channel.

This resolves the paradox without firewalls, complementarity, or modifications to quantum mechanics. The Hawking temperature acquires a correction:

$$T_{H,\text{UST}} = \frac{T_H}{N_b} \quad S_{\text{UST}} = N_b^2 S_{\text{BH}} \quad (16)$$

5 Cyclic Universe: Big Bang as Channel Inversion (T25)

Theorem 6 (T25 – Exact Conservation Law).

$$T_{\Omega}(N_b, C_{cb}) = T_{\Omega}(C_{cb}, N_b) = 0.23252885 \quad (exact) \quad (17)$$

The universe oscillates between two states:

$$\text{Odd cycles: } N_b = 0.63354460 \quad (\text{our universe}) \quad (18)$$

$$\text{Even cycles: } N_b = C_{cb} = 0.36645540 \quad (19)$$

Cycle frequency: $1/T_{\Omega} = 4.3005$. Dark matter and dark energy are the frozen gravitational remnants of the previous cycle's active physics. The Big Bang is not a singularity – it is a phase transition.

6 Falsification Boundary (Popper Criterion)

UST makes two parameter-free predictions testable by 2028:

1. **ESA Euclid / CMB-S4:** $\Omega_{\Lambda} = 0.6857$ at $<0.5\%$ precision.
2. **CMB-S4 polarisation:** $\chi_{\text{pol}} = \pi/8 = 22.500^{\circ}$ at the Omnium horizon.

If either measurement deviates significantly, UST must be discarded. No parameter adjustment is possible.

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